COMMERCIAL ITEM DESCRIPTION

ELECTROCARDIOGRAPH (ECG)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a non-transmission type, multichannel, digital electrocardiograph (ECG).

Salient characteristics:

1. Unit shall be a multichannel electrocardiograph which:

a. Acquires standard 12-lead electrocardiogram (I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6).

b. Digitizes the information.

c. Performs algorithmic analysis, such as filtering noise, detecting arrhythmia abnormalities, etc.

d. Provides a printed, three channel electrocardiogram with narrative interpretation.

e. Allows for patient data entry through a full, alpha-numeric keyboard.

2. Alphanumeric keyboard shall be capable of entering and recording multiple patient data entries. Data shall include but is not limited to: patient name, social security number, age, height, weight, sex, medication and patient status, i.e. inpatient or outpatient. Time and date of ECG and patient's name and social security number shall be annotated on the hard copy.

3. Integral alphanumeric printer shall provide multiple recording formats and at least three channel (50 mm) records with patient information and ECG interpretation. The unit shall be microprocessor controlled with operator selectable speeds and sensitivities of at least 25 and 50 mm/sec and 5, 10 and 20 mm/mV with an accuracy of plus or minus 5 percent, respectively.

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4. Unit shall provide immediate analysis of patient ECG at the bedside by the unit's internal microprocessor vice using a central computer in the absence of a medical officer. Hard copy shall be automatically annotated with a statement "Unconfirmed Report" to alert personnel to the limitations of interpretation validity.

5. Shall include mobile cart and internal memory storage for a minimum of fifty (50) annotated 10 sec, 12 lead ECG's. Digital tape storage is permissible.

6. The mobile cart shall be designed to accommodate all equipment required for a fully operational system.

7. Data furnished shall include patient information, accurate parameter measurements, morphological interpretation and rhythm analysis. There shall be an area for operator editorial or text input. Text shall be entered via the keyboard at the cart. Hard copy shall contain a statement that the analysis is unconfirmed if editing confirmation has not occurred. Unit shall be capable of at least 3 lead simultaneous disclosure of a 10 second ECG. Lead selection shall be operator selectable from 3 to 12 leads with access to the following lead groups: bipolar, unipolar extremity and unipolar precordial. All leads shall be clearly anatomically labeled.

8. Integral software shall generate annotated physiological parameters acquired from the analysis of raw data and ultimately generate a feasible morphological interpretation. Physiological parameters identified shall be at least heart rate, PR interval, QRS duration, QT and QT corrected and electrical axis measurements. Morphological interpretation shall encompass conduction blocks, ventricular hypertrophies, infarctions, ST and T wave abnormalities, atrial enlargement, premature beats, supraventricular tachyarrhythmias, ventricular tachyarrhythmias, bradyarrhythmias, junctional arrhythmias, ventricular pre-excitation, and facilitate discernment of pulmonary disease and drug effects.

9. Unit shall be defibrillator protected to at least 360 joules.

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10. The unit shall include the following functions:

a. Arm lead reversal indicator.
b. Lead-off and artifact detector.
c. Drift reduction.
d. AC interference rejection.

11. Electrical Requirements:

a. Shall be capable of operating over the full tolerance arrange of one or more of the following combinations of line voltages and frequency combinations as specified by the customer requisition:

1. 115 VAC plus or minus 10 percent, 50 plus or minus 2Hz
2. 115 VAC plus or minus 10 percent, 60 plus or minus 2Hz
3. 230 VAC plus or minus 10 percent, 50 plus or minus 2Hz
4. 230 VAC plus or minus 10 percent, 60 plus or minus 2Hz

A dual voltage and hertz feature on all units is desirable but not mandatory.

b. Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

12. Unit shall meet all applicable requirements of the following:

a. ANSI/AAMI EC 11, Standard for Diagnostic Electrocardiographic Devices.

b. UL544, Standard for Safety for Electric Medical and Dental Equipment.

c. AHA publication "Recommendations for Standardization of Leads and of Specifications for Instruments in Electrocardiography and Vectorcardiography".

d. AAMI ECGC, Standard for ECG Connectors.

13. System shall have a frequency response that is flat between 0.14 Hz to 50 Hz with no more than a 3db drop at 0.05 Hz and 100 Hz.

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14. Input impedance-differential shall be greater than 5 megohms, 0.5 Hz to 100 Hz.

15. Common mode rejection ratio shall be at least 10,000:1 (80 db) for a common mode signal of 100 mV.

16. Appropriate filtering and shielding shall be incorporated to minimize radio frequency signal interference.

17. Patient leads shall be isolated from ground for all line-operated units and all battery operated units that can also be line powered. The risk current flowing to or from the patient electrode connections, chassis, or controls of the electrocardiograph shall not exceed the limits specified in ANSI/AAMI ES1-1985, Safe Current Limits for Electromedical Apparatus.

18. ECG recordings shall be easily reproducible on all types of office photocopiers.

19. System shall be portable.

20. Operation instructions shall be clearly mounted on the unit.

21. Patient cable and leads shall be at least 1.5 m long.

22. Auto lead switching with option for manual lead selection shall be provided.

23. Unit shall permit operator selectability of lead or lead group during recording.

24. The unit shall have the capability of operator selected 12-lead ECG recording and simultaneous 3-lead rhythm strip recording of the following lead groups: I, II, III; aVR, aVL, aVF; V1, V2, V3, V4, V5, and V6, without activating the alphanumeric keyboard and internal microprocessor.

25. Shall have a high speed micro-processor controlled thermal printer.

26. The unit shall accurately mark all leads, speeds and sensitivities on the hard copy record of the grid paper. Hard copy of a standard 4 lead group recording shall be automatically presented on 8-1/2 x 11 inch grid paper. Roll paper with integral cutter is acceptable. Presentation of the ECG in a standard 8-1/2 x 11 inch format shall not require paper manipulation such as mounting or pasting.

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27. Unit shall provide recording at variable speeds to compress ECG recording. Rhythm strip recording speeds shall be available at the operator option at nominal speeds of at least 25 and 50 mm/sec, accurate to plus or minus 5 percent for time intervals between 0.2 and 2.0 seconds.

28. Unit shall permit the tracking of time from the beginning of each recording. Time shall be annotated when a recording is initiated. Time of ECG acquisition shall be annotated and invariable, not subject to deletion.

29. Unit shall provide standardization signal to be injected automatically in all leads for instrument calibration check.

30. Linearity shall be plus or minus 5 percent (plus or minus 10 mV).

31. Decay time (time constant) shall be such that 3mm or more shall remain from an initial 10mm deflection after an elapsed time of 3.2 seconds. Initial decay shall not exceed 5 percent in the first 0.2 seconds.

32. The unit shall be capable of displaying an ECG signal in the presence of pacemaker pulses with amplitudes between 2 and 250 mV, duration of 0.1 to 2.0 msec, a rise time of less than 100 microseconds, and a frequency of 100 pulse/min. For pacemaker pulses having durations between 0.5 and 2.0 msec, an indication of the pacemaker pulse shall be visible on the recording; this indication on the display shall have an amplitude of at least 0.2 mV.

33. Unit shall be supplied complete with all required interfacing and auxilliary equipment, accessories and disposable consumables necessary for immediate, full operation of bedside analysis.

34. The unit shall be of a rugged construction capable of withstanding hazards of normal use in a hospital setting.

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35. Operation and Service Data Manuals:

a. Information Content:

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematics and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc, pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

b. Quantities, Distribution and Schedules:

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of 2 service manuals need to be supplied to each consignee. However, 2 operation manuals still need to be supplied with each unit.

Workmanship. The electrocardiograph shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electrocardiograph, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the electrocardiograph (ECG) when a first article sample is required. This inspection shall include examination and tests.

Sampling For tests. Sampling for tests shall be conducted in accordance with commercial practice. The unit of product for test purposes shall be one electrocardiograph.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Performance Test. The electrocardiograph shall be tested for compliance with requirements as follows:

a. Using an electronic patient simulator, a series of known normal and abnormal ECG's shall be used to determine the unit's ability to recognize, display, and store the data correctly. Storage of interpretation/analysis, waveform data and patient information will be examined. Correct morphological interpretation of the abnormalities listed under Salient Characteristics, paragraph 8, shall be verified.

b. The unit's ability to correctly annotate all leads shall be verified. Lead switching, the availability and implementation of multiple formats and undistorted memory retrieval shall be examined to verify that the unit complies with contractor's claims and government specification.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

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6515-01-305-1160

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-4753

Location: CID/ECG/H16-22

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-51945	ELECTROCARDIOGRAPH (ECG)

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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